

Catherine Chen

SUMMARY OF SKILLS AND QUALIFICATIONS

- 1+ year in cross-functional co-op teams covering enterprise risk, procurement, analytics, and process improvement
- Skilled in financial analysis, quantitative risk, statistics, and ML, with communication, problem-solving, and multitasking abilities
- Proficient in Python, R, SQL, MATLAB, Google Analytics, Excel/VBA, and Power BI for data analysis and automation

EDUCATION

- **Candidate, Master of Financial Insurance (MFI)** Sep. 2026 - Present (Excepted completion 2027)
University of Toronto, Department of Statistical Sciences
Relevant Courses: Mathematical Finance, Insurance Risk Management, Data Science and Machine Learning
- **Bachelor of Mathematics, Co-op Program (With Distinction)** Sep. 2021 - Jun. 2026
University of Waterloo, Department of Statistics and Actuarial Science
Honours Mathematics/Financial Analysis and Risk Management (FARM)
- **Candidate for the FRM Certification (Level I)**

PROFESSIONAL EXPERIENCES

- **Teaching Assistant, University of Toronto** Sep. 2025 – Present
Toronto, Canada
 - Coordinated grading to maintain consistent assessment standards and proctored exams
 - Converted solution files to **Quarto (.qmd)** for consistent formatting and answered questions to clarify course concepts
- **Procurement & Strategic Operations (Co-op), Municipal Property Assessment Corporation (MPAC)** Apr. 2025 – Aug. 2025
Pickering, Canada
 - Developed an insurance compliance dashboard for **120+ suppliers**, automating expiry alerts and cutting follow-ups by **75%**
 - Co-led a workflow survey across **6 business units**, using descriptive analytics to identify bottlenecks and improve collaboration
 - Supported **supplier risk models** from EDIA and Code of Conduct disclosures, creating a dataset for heatmapping and tracking
- **Insurance and Enterprise Risk (Co-op), Toronto District School Board (TDSB)** May 2024 – Dec. 2024
North York, Canada
 - Automated tracking for **200+ vendors** with **Excel formulas**, reducing manual work across onboarding, renewals, and approvals
 - Built **Excel VBA** to classify email inquiries and summarize recurring issues, improving response efficiency and transparency
 - Maintained the Excursion website using **HTML/CSS**, resolving technical issues and redesigning layout to improve usability
- **Web Assistant (Co-op), University of Waterloo** Sep. 2022 – Dec. 2022 & May 2023 – Aug. 2023
Waterloo, Canada
 - Modernized **100+ pages** (**Work-Learn Institute**), including **link validation**, **layout redesign**, and bug fixes using **HTML/CSS**
 - Built responsive pages from wireframes and refined layouts for **Centre for Work-Integrated Learning** in a 6-person team
 - Developed **Python** for data collection and **Google Analytics** for engagement analysis, improving web maintenance efficiency

RESEARCH & TECHNICAL PROJECTS

- **Two-Factor Analysis of Workout Type and Time on Calorie Burn** Jan. 2025 - Apr. 2025
Waterloo, Canada
 - Conducted a **two-factor randomized complete block experiment** on calorie burn with 12 participants and 6 treatments
 - Collected and cleaned experimental data; implemented **ANOVA and visualizations in R** to assess main and interaction effects
 - Interpreted **p-values and interaction plots**: workout type dominated, while time of day had a smaller significant effect
- **Machine Learning-Based Evaluation of Small and Micro Business Projects, East China Normal University** Sep. 2023 – Dec. 2023
Shanghai, China
 - Encoded expert reviews from **314 project evaluation articles** into structured numerical data across six dimensions
 - Developed an **MLP model in R (RSNNS)** to predict investment recommendations, achieving **81.8% validation accuracy**
 - Conducted **data cleaning**, manual annotation, outlier filtering, and train-validation splitting to ensure model robustness

EXTRACURRICULAR EXPERIENCES

- **Content Creator, Bilibili (China's YouTube-equivalent)** May 2021 – Present
Produced **30+ original videos**, earning **9,000+ likes** on one video and demonstrating strong audience engagement